

NTT-Friendly Pairing Curve Construction with Readability-Aware Prime Selection over a 1024-bit Field

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Abstract—Pairing-friendly elliptic curves are fundamental to advanced cryptographic protocols such as BLS signatures and zero-knowledge proof systems. However, the traditional Cocks-Pinch construction produces primes with no guaranteed algebraic structure, limiting their utility for NTT-based proof systems and hindering parameter auditability. We present three contributions: (1) a modified Cocks-Pinch algorithm that constrains the cyclotomic seed to guarantee two-adicity of at least 36 in both the base field and scalar field primes, exceeding the industry-standard BLS12-381 curve; (2) a multi-criteria readability scoring system that selects human-auditable primes from the valid candidate space without weakening security; and (3) Curve1024, a concrete 1024-bit pairing-friendly curve ($E : y^2 = x^3 + 41$, $k = 18$, 256-bit ECDLP security) with full parameter set and generator point published for public verification. A complete Rust implementation with 36 passing tests demonstrates practical signing operations and validates resistance against MOV, Anomalous, and TNFS attacks.

Index Terms—elliptic curve cryptography, pairing-friendly curves, Cocks-Pinch method, NTT-friendly primes, parameter auditability

I. INTRODUCTION

Pairing-friendly elliptic curves underpin a growing class of cryptographic protocols, from BLS aggregate signatures [1] to succinct non-interactive zero-knowledge proofs (zk-SNARKs) [2], [3]. These protocols require curves whose scalar field and base field support efficient polynomial arithmetic via the Number Theoretic Transform (NTT), which in turn demands that both field primes satisfy $q \equiv 1 \pmod{2^s}$ for a sufficiently large s .

The Cocks-Pinch method [4] offers unmatched flexibility in constructing pairing-friendly curves with arbitrary embedding degrees. Unlike parametric families such as BN [5] or BLS12 [6], the Cocks-Pinch approach can target specific security levels and field sizes. However, the traditional algorithm treats NTT-friendliness as an afterthought—primes are generated with random two-adicity, typically only 1–3 bits, rendering them unsuitable for modern proof systems.

A separate but related concern is *parameter auditability*. The Dual_EC_DRBG incident [7] demonstrated that opaque cryptographic constants can conceal backdoors. While the Cocks-Pinch method is inherently transparent (all parameters derive from a public seed), the resulting primes are typically inscrutable hexadecimal blobs that resist manual inspection.

Contributions. This paper presents three results:

1. **Improved Cocks-Pinch with NTT constraints.** We modify the Cocks-Pinch construction to *guarantee* high two-adicity in both field primes by constraining the cyclotomic seed T . Choosing $T \equiv 0 \pmod{2^{12}}$ yields $\nu_2(r-1) \geq 36$ deterministically, while an extended lift search ensures $\nu_2(p-1) \geq 36$. The resulting two-adicity of 36 exceeds BLS12-381’s value of 32.
2. **Readability-aware prime selection.** We introduce a multi-criteria scoring system that evaluates candidate primes on eight structural properties (binary sparseness, zero limbs, hex density, etc.) and selects the most human-auditable parameters without sacrificing security.
3. **Curve1024: a concrete 1024-bit pairing-friendly curve.** We instantiate the above techniques to produce a complete, publicly verifiable curve with embedding degree $k = 18$, scalar field order $r \approx 2^{512}$ (256-bit ECDLP security), and equation $E : y^2 = x^3 + 41$. The full parameter set—including generator point—is published and validated by a Rust implementation with 36 passing tests.

Organization. Section 2 reviews background on pairing-friendly curves and the CM method. Section 3 presents our NTT-friendly Cocks-Pinch modification. Section 4 describes the readability scoring system. Section 5 presents the Curve1024 parameters and evaluates the implementation. Section 6 discusses related work, and Section 7 concludes.

II. PRELIMINARIES

A. Elliptic Curves and Pairings

Let $\mathbb{F}_p$ be a prime field with $p > 3$. An elliptic curve in Short Weierstrass form is $E : y^2 = x^3 + ax + b$ with $4a^3 + 27b^2 \neq 0$. The set $E(\mathbb{F}_p)$ of rational points forms an abelian group under the chord-and-tangent law [8].

A curve is *pairing-friendly* with embedding degree k if there exists a prime-order subgroup of size r such that $r \mid p^k - 1$ but $r \nmid p^i - 1$ for $0 < i < k$. This enables a bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ where $\mathbb{G}_T \subset \mathbb{F}_{p^k}^*$, which is the foundation of BLS signatures and zk-SNARKs.

B. Complex Multiplication Method

The CM method constructs curves with a prescribed group order. Given a CM discriminant $D < 0$, the existence of a curve over $\mathbb{F}_p$ with Frobenius trace t requires the Diophantine equation:

$$4p = t^2 + |D| \cdot y^2$$

to have an integer solution. The j -invariant is obtained from the Hilbert class polynomial $H_D(x)$. For $D = -3$, we have $H_{-3}(x) = x$, giving $j = 0$ and the simple curve form $y^2 = x^3 + b$.

C. Traditional Cocks-Pinch Algorithm

The Cocks-Pinch method [4] constructs pairing-friendly curves by starting from the subgroup order r rather than the field prime p . For embedding degree $k = 18$:

1. **Generate r :** Choose a parameter T and compute $r = \Phi_{18}(T) = T^6 - T^3 + 1$. Verify that r is prime.
2. **Compute square root:** Find $\beta = \sqrt{-3} \pmod{r}$ via Tonelli-Shanks.
3. **Compute CM parameters:** For each i with $\gcd(i, 18) = 1$:

$$t_0 \equiv T^i + 1 \pmod{r}, \quad y_0 \equiv \frac{t_0 - 2}{\beta} \pmod{r}$$

4. **Lift to integers:** Search over $(h_t, h_y) \in [-20, 20]^2$ for:

$$t = t_0 + h_t \cdot r, \quad y = y_0 + h_y \cdot r$$

such that $p = (t^2 + 3y^2)/4$ is a prime of the target size.

The resulting curve has $\rho = \log p / \log r \approx 2$, which is the standard tradeoff of the Cocks-Pinch method compared to parametric families ($\rho = 1$ for BN, $\rho = 1.5$ for BLS12).

D. Two-Adicity and NTT

The *two-adicity* of a prime q is $\nu_2(q-1)$, the largest power of 2 dividing $q-1$. A field $\mathbb{F}_q$ with two-adicity s contains a primitive 2^s -th root of unity, enabling NTT of length up to 2^s . Modern ZK proof systems (Groth16 [2], PLONK [3]) require NTT for polynomial multiplication over both $\mathbb{F}_r$ (constraint evaluation) and $\mathbb{F}_p$ (pairing computation). BLS12-381 achieves two-adicity 32 in both fields [6].

III. IMPROVED COCKS-PINCH WITH NTT CONSTRAINTS

We modify the traditional Cocks-Pinch algorithm to guarantee high two-adicity in both output primes. The key insight is that the algebraic structure of the cyclotomic polynomial Φ_{18} allows deterministic control over the two-adicity of r , while an extended lift search achieves the same for p .

A. NTT-Friendly Scalar Field via Seed Constraint

Since $r = \Phi_{18}(T) = T^6 - T^3 + 1$, we observe:

$$r - 1 = T^6 - T^3 = T^3(T^3 - 1)$$

If $T \equiv 0 \pmod{2^m}$, then $T^3 \equiv 0 \pmod{2^{3m}}$. Since $T^3 - 1$ is odd (as T^3 is even), we obtain:

$$\nu_2(r - 1) = \nu_2(T^3) = 3m$$

Consequence: To guarantee $\nu_2(r - 1) \geq s$, it suffices to choose $T \equiv 0 \pmod{2^{\lceil s/3 \rceil}}$. This is a *deterministic* guarantee requiring no rejection sampling.

For our target of $s = 36$, we set $m = \lceil 36/3 \rceil = 12$, constraining $T \equiv 0 \pmod{2^{12}}$. The search space remains $2^{86-12} = 2^{74}$ candidates—more than sufficient for finding prime r .

B. NTT-Friendly Base Field via Extended Lift Search

After obtaining a valid r with the desired two-adicity, the base field prime is:

$$p = \frac{(t_0 + h_t r)^2 + 3(y_0 + h_y r)^2}{4}$$

The two-adicity of $p - 1$ depends on the lift parameters (h_t, h_y) . To ensure $\nu_2(p - 1) \geq s_p$, we extend the search grid beyond the default $[-20, 20]^2$:

$$\text{half_range} = 20 + 2^{\max(0, s_p - 3m)}$$

This extension guarantees sufficient coverage of bit patterns in $p - 1$ at positions above $3m$. For $s_p = 36$ and $m = 12$ (so $3m = 36$), the grid remains at $[-21, 21]^2$ —only marginally larger than the default.

C. Algorithm

The complete NTT-friendly Cocks-Pinch construction proceeds as follows:

Input: Embedding degree $k = 18$, target sizes $(|r|, |p|)$, minimum two-adicity s

Output: Primes (p, r) with $\nu_2(r - 1) \geq s$ and $\nu_2(p - 1) \geq s$

1. Set $m \leftarrow \lceil s/3 \rceil$ and step $\leftarrow 2^m$

2. Repeat:

- a. Choose T as a random multiple of step in the target range
- b. Compute $r \leftarrow T^6 - T^3 + 1$
- c. If r is not prime or $|r| \neq \text{target}$, continue
- d. Compute $\beta \leftarrow \sqrt{-3} \pmod{r}$ (Tonelli-Shanks)
- e. For each i with $\gcd(i, 18) = 1$:
 - $t_0 \leftarrow T^i + 1 \pmod{r}$
 - $y_0 \leftarrow (t_0 - 2) \cdot \beta^{-1} \pmod{r}$
 - $\text{range} \leftarrow 20 + 2^{\max(0, s - 3m)}$
 - For $(h_t, h_y) \in [-\text{range}, \text{range}]^2$:
 - $t \leftarrow t_0 + h_t \cdot r; y \leftarrow y_0 + h_y \cdot r$
 - $p \leftarrow (t^2 + 3y^2)/4$
 - If $|p| = \text{target}$ AND $\nu_2(p - 1) \geq s$ AND p is prime: return (p, r, t, y)

3. Until max attempts reached

TABLE I
TRADITIONAL VS. IMPROVED COCKS-PINCH

Metric	Traditional	Improved
Constraint on T	None	$T \equiv 0 \pmod{2^{12}}$
$\nu_2(r-1)$	Random (1–3)	≥ 36 (guaranteed)
$\nu_2(p-1)$	Random (1–3)	≥ 36 (with ext. grid)
Grid size	$[-20, 20]^2$	$[-21, 21]^2$
Avg. attempts	100–500	3,000–6,000
Generation time	2–10 s	30–60 s
NTT support	No	Up to 2^{36}

D. Complexity Analysis

The following table compares the traditional and improved algorithms. The NTT constraint increases the average number of attempts by approximately $10\text{--}60\times$ (from $\sim 100\text{--}500$ to $\sim 3,000\text{--}6,000$), but the absolute generation time remains under one minute—a one-time cost amortized over the lifetime of the curve parameters.

E. Comparison with BLS12-381

BLS12-381 [6], the current industry standard for ZK proof systems, achieves two-adicity 32 in both fields. Our construction achieves 36, enabling NTT of length $2^{36} \approx 6.8 \times 10^{10}$ —a $16\times$ improvement in maximum polynomial degree. The tradeoff is a larger ρ -value (2.0 vs. 1.5) and correspondingly larger field elements, which is acceptable for applications prioritizing long-term security (256-bit ECDLP vs. 128-bit for BLS12-381).

IV. READABILITY-AWARE PRIME SELECTION

A. Motivation

Cryptographic primes are typically opaque hexadecimal values that resist manual inspection. While the Cocks-Pinch method is inherently transparent—all parameters derive from a public seed T —the resulting primes still appear as random bit strings. This opacity complicates auditing: reviewers cannot easily verify structural properties or detect anomalies by inspection.

The Dual_EC_DRBG incident [7] demonstrated that even standardized constants can harbor backdoors when their generation process is opaque. We propose a complementary approach: among all cryptographically valid candidates, select the one that is most amenable to human inspection.

B. Security-First Design

Our readability scoring is a *post-filter* applied only to candidates that have already passed all security requirements:

1. Primality (Miller-Rabin with sufficient rounds)
2. Correct bit length (512 for r , 1024 for p)
3. Minimum two-adicity ($\nu_2(q-1) \geq 36$)
4. Valid CM equation ($4p = t^2 + 3y^2$)
5. Embedding degree verification ($r \mid \Phi_{18}(p)$)

The scoring system cannot weaken security because it only selects among pre-validated candidates.

TABLE II
READABILITY SCORING CRITERIA

Criterion	Rewards	Score Formula
Binary sparseness	Low Hamming weight	$(n - \text{hw}) \times 2$
Two-adicity	Higher $\nu_2(q-1)$	$\nu_2 \times 3$
Zero limbs	64-bit blocks = 0	count $\times 60$
Full limbs	Limbs = $0\text{xFF} \dots \text{F}$	count $\times 50$
Longest zero run	Contiguous zero limbs	run $\times 40$
Hex zero density	Hex digits = 0	$(z - 16) \times 2$
Simple top limb	Few bits in MSB limb	$(32 - \text{hw}) \times 3$
Repeating limbs	Adjacent equal limbs	pairs $\times 25$

TABLE III
READABILITY METRICS OF SELECTED PARAMETERS

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024	235/512
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16
Hex zero digits	33/256	16/128

C. Scoring Criteria

Each candidate prime is evaluated on eight independent structural criteria:

The combined score weights the base field prime p at $1.5\times$ the scalar field prime r , since p appears more frequently in implementations (field arithmetic, point encoding, serialization).

D. Search Procedure

The readability-aware search operates in two modes:

- **Fast mode:** Returns the first valid (p, r) pair (original behavior).
- **Readable mode:** For each valid r , explores the *full* lift range to find the highest-scoring p . Tracks the global best across N successful attempts (default $N = 50$) within a time budget.

In readable mode, the generator exhaustively searches all (h_t, h_y) combinations for each valid r , scoring every prime candidate that passes the security filter. The best-scoring (p, r) pair across all attempts is retained.

E. Results

The following table shows the readability metrics of our selected parameters. Both primes share the tail pattern $\dots \text{d970_00000001}$, a consequence of $T \equiv 0 \pmod{2^{12}}$ propagating through the construction. This shared structure aids visual verification.

The scalar field prime r achieves 8 zero limbs out of 16 (50%), meaning half of its 64-bit representation is zero—a property that simplifies implementation and testing. The shared tail $\text{d970_00000001} = d_{970} \cdot 2^{36} + 1$ visually confirms the NTT-friendly structure.

V. CURVE1024: PARAMETERS AND EVALUATION

This section presents the concrete Curve1024 parameters produced by our improved Cocks-Pinch construction with

TABLE IV
CURVE1024 SUMMARY

Parameter	Value
Equation	$E : y^2 = x^3 + 41$
Embedding degree	$k = 18$
CM discriminant	$D = -3$ ($j = 0$)
$\rho = \log p / \log r$	2.0

readability-aware selection, followed by implementation details and security analysis.

A. Curve1024 Parameters

The construction produces the following pairing-friendly elliptic curve over a 1024-bit prime field. All parameters are publicly verifiable from the seed T .

1) *Seed and Construction:* **Seed:** $T = 0x26704d2ace0facdd539 \cdot 2^{12}$

The seed satisfies $T \equiv 0 \pmod{2^{12}}$, guaranteeing $\nu_2(r - 1) = 36$ via the relation $r - 1 = T^3(T^3 - 1)$.

2) *Scalar Field Order* ($r = \Phi_{18}(T) = T^6 - T^3 + 1$, 512 bits):

```
r = 0x c042c338 9e72044d 9ec8078a ea7bc954
      ebf209d9 18f0ef27 425259e2 47711927
      97463bf2 3832d0de 5e46e47f c61cab50
      4bcfd36f c2a945c5 1055d970 00000001
```

Properties: $|r| = 512$ bits, $\nu_2(r - 1) = 36$, 8 zero 64-bit limbs out of 16.

3) *Base Field Prime* ($p = (t^2 + 3y^2)/4$, 1024 bits):

```
p = 0x c0859da8 3a500988 10eea35c 61fea054
      c8669025 99220d30 c93f3826 5ed6b830
      5785b8a0 73e482e8 685cfeeb 388dc45e
      bca16862 45b326fe 824ca74a c0844160
      f4e28980 6084e50e 6c02f32c b94d786b
      f80fb080 c07d6a07 73f2b459 1eb70df3
      9367aa32 ebb42449 e0a648f6 1a20e876
      f971f3bf 976a7d6a dd20d970 00000001
```

Properties: $|p| = 1024$ bits, $\nu_2(p - 1) = 36$, Hamming weight 464/1024.

4) *Curve Equation and Generator:* **Generator point** $G = (G_x, G_y)$ of order r :

```
Gx = 0x 0c04c9bd 75d883ea 501c1b00 33b81dac
      69a688eb 67bd7126 0fc49d36 6674a0c0
      4686227983a7188a ed3767b9 ff894eeb
      2cdf8073 5a9a4456 afbfe264 e8139ad5
      b0e90ca9 a9588464 771871c2 d8b86b09
      a52440ff e1b9984e 1566ace7 87feea0d
      11310eec baf0a39d 0d3feb5b 7279c384
      d74dc993 caef5866 7681783133a11cfa
```

```
Gy = 0x 9dde8a42 a8a618aa c7691f9d e4d4e626
      d037825e 33d925f0 775be9a8 d60969c9
      4fe1033b a71c6518 5280e262 11a43ddf
      30bd4d40 60d21a65 b5234cd9 f29763c2
```

TABLE V
READABILITY METRICS

Metric	p (1024-bit)	r (512-bit)
Hamming weight	464/1024	235/512
Two-adicity	36	36
Zero 64-bit limbs	0/16	8/16
Hex zero digits	33/256	16/128

TABLE VI
PERFORMANCE BENCHMARKS

Operation	Min	Mean	Max
Key generation	1.629 s	1.687 s	1.757 s
Schnorr sign	1.473 s	1.551 s	1.641 s
Schnorr verify	2.656 s	2.831 s	3.025 s
ECDSA sign	1.524 s	1.606 s	1.711 s
ECDSA verify	2.847 s	3.140 s	3.459 s
Scalar mult. $[k]G$	1.061 s	~1.2 s	1.489 s

```
b70048d7 fdd4a9be f0cc4c70 1bbed367
8a371139 fcadacc5 b8dc9361 469d8ddc
c5c93289 767dceca 4763febd 6a618277
2dcf7bfc 3dc1cce3 d8c75862 4a40c468
```

5) *Structural Properties:* Both primes share the tail pattern $\dots d970_00000001$, visually confirming the NTT-friendly structure $q = d \cdot 2^{36} + 1$. This shared suffix is a direct consequence of the seed constraint $T \equiv 0 \pmod{2^{12}}$ propagating through the cyclotomic polynomial.

B. Implementation

We implement the complete system in Rust without external cryptographic dependencies [9]. The implementation comprises:

- A custom U1024 big integer type with Montgomery multiplication [10]
- Prime field arithmetic (PrimeFieldElement) with constant-time Fermat inversion
- Elliptic curve group operations in affine coordinates
- Schnorr and ECDSA signature schemes
- The parameter generation tool (improved Cocks-Pinch + readability scoring)

The test suite contains 36 tests covering arithmetic correctness, group law verification, and attack simulation (MOV, Anomalous, Invalid Curve).

C. Performance Benchmarks

The following table reports timing measurements using the Criterion framework (release build, LLVM O3 optimization, 10 samples per measurement, commodity hardware without AVX-512).

The performance reflects the 1024-bit field size: each scalar multiplication requires ~512 double-and-add iterations, each involving a Fermat inversion (~1536 Montgomery multiplications). This is approximately 8000 $\times$ slower than optimized secp256k1 implementations, which is expected given the 4 $\times$ larger field and $O(n^2)$ multiplication cost. The system targets

TABLE VII
SECURITY EVALUATION SUMMARY

Attack	Complexity	Status
Pollard- ρ (ECDLP)	$O(2^{256})$	Secure
MOV reduction	DLP in $\mathbb{F}_{p^{18}}$ (18432-bit)	Secure
Anomalous (SSSA)	Requires $\#E = p$	Immune
TNFS	≥ 250 -bit in $\mathbb{F}_{p^{18}}$	Secure
Invalid curve	Point validation in constructor	Mitigated

TABLE VIII
COMPARISON WITH STANDARD CURVE FAMILIES

Curve	$ p $	ρ	ν_2	Security
BN254	254	1.0	~1	100-bit
BLS12-381	381	1.5	32	128-bit
KSS18 (generic)	varies	1.33	random	varies
Curve1024	1024	2.0	36	256-bit

offline signing scenarios (legal documents, software releases, PKI certificates) where seconds of latency are acceptable.

D. Security Analysis

The 256-bit ECDLP security exceeds NIST SP 800-57 recommendations through 2030+ [11]. The MOV attack maps ECDLP to DLP in $\mathbb{F}_{p^{18}}$, a field of 18,432 bits—far exceeding the 3,072-bit threshold for 128-bit security. The curve is provably non-anomalous since $\#E(\mathbb{F}_p) = h \cdot r$ with $r \approx 2^{512} \neq p \approx 2^{1024}$.

Limitation: The current affine-coordinate implementation uses variable-time double-and-add, which leaks scalar bits through timing. A Montgomery ladder (constant-time) is identified as future work.

E. Curve Family Comparison

VI. RELATED WORK

Pairing-friendly curve taxonomy. Freeman, Scott, and Teske [5] provide a comprehensive classification of pairing-friendly curve constructions, including the Cocks-Pinch method, MNT curves, and polynomial families (BN, BLS, KSS). Their work establishes the theoretical framework but does not address NTT-friendliness as a design criterion.

TNFS security updates. Barbulescu and Duquesne [12] demonstrated that the Tower Number Field Sieve reduces the effective security of many pairing-friendly curves, notably downgrading BN256 from 128-bit to approximately 100-bit security. This motivates our choice of a 1024-bit base field with $k = 18$, yielding a pairing target of 18,432 bits.

BLS12-381. Bowe [6] designed BLS12-381 specifically for zk-SNARKs, achieving two-adicity 32 in both fields through careful parameter selection within the BLS12 family. Our work achieves higher two-adicity (36) using the more flexible Cocks-Pinch method, at the cost of a larger ρ -value.

Pasta curves. The Pallas/Vesta cycle [13] achieves NTT-friendliness through a 2-cycle of curves, enabling recursive proof composition. Unlike our approach, Pasta curves are

not pairing-friendly and target a different application domain (Halo 2 recursive proofs).

Nothing-up-my-sleeve numbers. The cryptographic community has long advocated for “nothing-up-my-sleeve” constants (e.g., using digits of π) to demonstrate absence of backdoors. Our readability scoring takes a complementary approach: rather than deriving primes from a fixed public constant, we select the most inspectable prime from a large space of equally secure candidates. Both approaches serve the goal of transparent parameter generation.

Cocks-Pinch extensions. Chen, Hong, and Wang [14] extend the Cocks-Pinch method to embedding degrees 5–8 with optimal Ate pairing computation. Our work focuses on $k = 18$ with the novel addition of NTT constraints and readability scoring.

VII. CONCLUSION

We presented three contributions to pairing-friendly elliptic curve construction:

1. **NTT-friendly Cocks-Pinch.** By constraining the cyclotomic seed $T \equiv 0 \pmod{2^{12}}$, we guarantee two-adicity ≥ 36 in the scalar field deterministically, while an extended lift search achieves the same for the base field. This exceeds the industry-standard BLS12-381 (two-adicity 32) at a modest increase in generation time (30–60 s vs. 2–10 s).
2. **Readability-aware prime selection.** Our multi-criteria scoring system produces primes that are more amenable to human inspection without any security degradation, addressing the auditability concern in cryptographic parameter generation.
3. **Curve1024.** We instantiate these techniques into a concrete, publicly verifiable curve: $E : y^2 = x^3 + 41$ over a 1024-bit prime field with embedding degree 18, providing 256-bit ECDLP security and NTT support up to length 2^{36} in both fields. The complete parameter set (field primes, generator point, CM data) is published and validated by a Rust implementation with 36 passing tests.

The system is suitable for offline signing applications requiring long-term security guarantees and for future pairing-based protocols (BLS aggregate signatures, zk-SNARKs) that benefit from the NTT-friendly field structure.

Future work includes: (1) projective/Jacobian coordinates to eliminate per-operation inversions, (2) constant-time Montgomery ladder for side-channel resistance, (3) full optimal Ate pairing implementation over $\mathbb{F}_{p^{18}}$, and (4) extension to other embedding degrees ($k = 12, 24$).

A. Availability

The implementation and all curve parameters are available as open-source software under the MIT license [9].

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